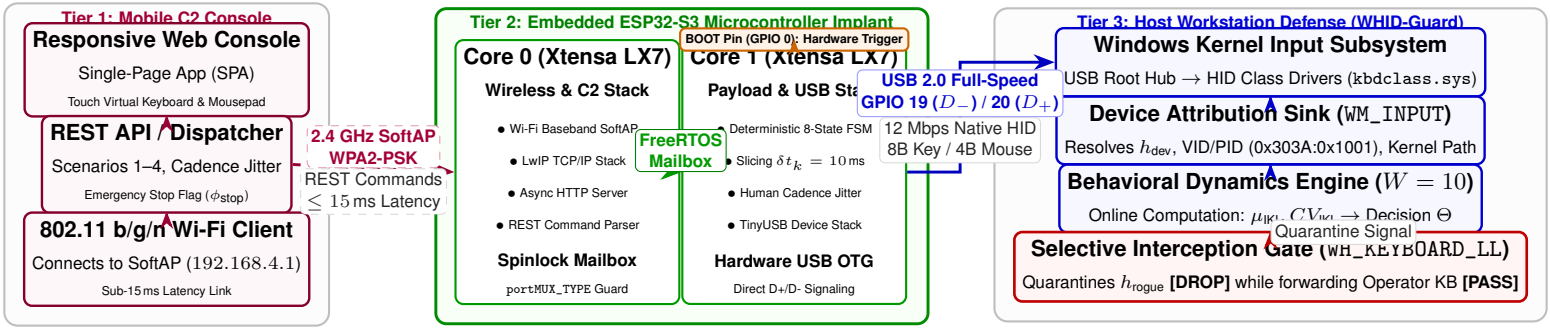

Tier 1: Mobile C2 Console
Responsive Web Console
Single-Page App (SPA)
Touch Virtual Keyboard & Mousepad
REST API / Dispatcher
Scenarios 1–4, Cadence Jitter
Emergency Stop Flag (ϕ_{stop})
802.11 b/g/n Wi-Fi Client
Connects to SoftAP (192.168.4.1)
Sub-15 ms Latency Link
2.4 GHz SoftAP
WPA2-PSK
REST Commands
≤ 15 ms Latency
Tier 2: Embedded ESP32-S3 Microcontroller Implant
BOOT Pin (GPIO 0): Hardware Trigger
Core 0 (Xtensa LX7)
Wireless & C2 Stack
Wi-Fi Baseband SoftAP
LwIP TCP/IP Stack
Async HTTP Server
REST Command Parser
Spinlock Mailbox
portMUX_TYPE Guard
FreeRTOS
Mailbox
Core 1 (Xtensa LX7)
Payload & USB Sta
Deterministic 8-State FSM
Slicing $\delta t_k = 10$ ms
Human Cadence Jitter
TinyUSB Device Stack
Hardware USB OTG
Direct D+/D- Signaling
USB 2.0 Full-Speed
GPIO 19 (D_-) / 20 (D_+)
12 Mbps Native HID
8B Key / 4B Mouse
Tier 3: Host Workstation Defense (WHID-Guard)
Windows Kernel Input Subsystem
USB Root Hub → HID Class Drivers (kbdclass.sys)
Device Attribution Sink (WM_INPUT)
Resolves h_{dev}, VID/PID (0x303A:0x1001), Kernel Path
Behavioral Dynamics Engine ($W = 10$)
Online Computation: $\mu_{IKI}, CV_{IKI} \rightarrow$ Decision Θ
Quarantine Signal
Selective Interception Gate (WH_KEYBOARD_LL)
Quarantines h_{rogue} [DROP] while forwarding Operator KB [PASS]